

Universitas Pandrosion

A Formally Verified Multi-Start Steffensen Algorithm for the Complex Equation $z^p = x$

Machine-checked in Lean 4 / Mathlib v4.7.0, zero `sorry` declarations in the main spine

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github.com/ivan-fr/poussiere_de_x

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Abstract

We present a formally verified Lean 4 / Mathlib v4.7.0 development of the *Pandrosion-Steffensen algorithm* for solving the complex equation $z^p = x$ with $x > 1$. The core of the method is the rational iteration $h_{p,x}(z) = 1 - (x - 1)/(x \cdot S_p(z))$, where S_p is the geometric sum $\sum_{k=0}^{p-1} z^k$; its fixed points are exactly the p -th roots of $1/x$. Composed with the Aitken-Steffensen Δ^2 accelerator, the method becomes derivative-free and of quadratic order.

The formalization consists of 15 Lean modules in `Pandrosion.Core.*` that build, in order: (i) the geometric-sum identity $S_p(s)(1 - s) = 1 - s^p$ and the fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (real and complex); (ii) the multi-start Pandrosion-Steffensen algorithm with a Voronoï nearest-anchor selector; (iii) a constructive McMullen global-coverage theorem establishing that the selector uniquely assigns every $z_0 \in \mathbb{C}$ off a measure-zero Voronoï boundary; (iv) the cyclotomic anchor family $\gamma_k = \alpha \cdot e^{2\pi i k/p}$, whose anchors are pairwise distinct roots of $1/x$ and fixed points of the Steffensen iterator; (v) a closed-form linear contraction rate $\lambda_{p,x} = 1 - p \cdot \alpha^{p-1}/S_p(\alpha)$ together with a p -uniform complexity bound; (vi) Kung-Traub optimality for the derivative-free order-2 efficiency index; (vii) the complex multiplier $h'(\alpha)$ in closed form; (viii) local dynamical attraction on a disk around each anchor; and (ix) an almost-everywhere dynamical convergence theorem in the Lebesgue sense.

The 15-module spine compiles with zero `sorry` in the main theorems and is stitched by a single master-theorem statement, `master_absolu`, that asserts all five load-bearing pillars simultaneously. Where analytic content is deferred (the local super-attracting quadratic bound; global trajectory entry into the basin), hypotheses are stated explicitly so the reader can follow which pieces are unconditional and which are conditional.

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1 Introduction

1.1 The Pandrosion iteration

Given a positive real x and a natural $p \geq 1$, the *Pandrosion iteration* is the rational map

$$h_{p,x}(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \quad S_p(s) = \sum_{k=0}^{p-1} s^k. \quad (1)$$

It is defined on $\{s : S_p(s) \neq 0\}$ (both over $\mathbb{R}$ and over $\mathbb{C}$) and is derivative-free: no symbolic differentiation is required at evaluation time. The historical namesake honours Pappus of Alexandria's record of Pandrosion's geometric cube-duplication construction [3]; structurally, the map is reminiscent of Householder-type root-finders [6, 7] but is algebraically distinct from Newton and Halley.

The central algebraic fact is:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Thus the fixed points of (1) are exactly the p -th roots of $1/x$. For $x > 1$ this is p distinct complex numbers on the circle $|s| = x^{-1/p} < 1$. The Pandrosion-Steffensen algorithm is this map composed with the Aitken Δ^2 accelerator of Aitken [4] and Steffensen [5]; the accelerator turns the linear contraction of (1) into quadratic convergence without any derivative, and attains the Kung-Traub efficiency bound [8] for derivative-free methods with two evaluations per step.

1.2 What this paper proves

The paper documents a **15-module Lean 4 formalization** living in `Pandrosion/Core/*`, built against Mathlib v4.7.0, with zero `sorry` in the main spine. The narrative follows the Lean module order exactly:

1. **Algebraic foundations** (§2): geometric-sum identities, fixed-point characterisation (real and complex), Pandrosion-Steffensen accelerator, scaling invariance.
2. **Multi-start architecture** (§3): Voronoï half-plane basins, multi-start grand-master theorem, constructive McMullen global coverage, Voronoï boundary is Lebesgue-null, cyclotomic anchor family, unconditional McMullen.
3. **Rates and complexity** (§4): closed-form linear rate $\lambda_{p,x}$, Aitken-Steffensen quadratic-rate skeleton, p -uniform two-phase complexity bound, Kung-Traub efficiency optimality.
4. **Complex dynamics** (§5): complex multiplier at the fixed point in closed form, local dynamical attraction around each anchor (conditional on a local quadratic bound), almost-everywhere dynamical convergence (crown jewel).
5. **Master Absolu** (§6): the top-level stitching theorem combining five load-bearing pillars.

1.3 What this paper does *not* claim

Two pieces of genuinely analytic content are *deferred* and taken as explicit hypotheses in the relevant theorems:

- A **local quadratic bound** $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on a neighbourhood of each cyclotomic anchor γ_k , where σ is the Steffensen iterator. This is the super-attracting content of Steffensen's method (the multiplier at a Steffensen fixed point vanishes); it is stated in the literature and used here as a hypothesis.

- An **almost-everywhere trajectory entry** statement: for a.e. $z_0 \in \mathbb{C}$, some iterate $\sigma^{k_0}(z_0)$ lies in the local disk of some anchor γ_k . This is the Fatou-type global dynamical content and is again used as a hypothesis.

Both hypotheses appear explicitly in the relevant theorem statements (§5.2, §5.3); no hidden axioms are smuggled into the master theorem. The *algebraic* and *measure-theoretic* spine from foundations through Kung-Traub optimality is unconditional.

2 Algebraic Foundations (Foundations.lean)

The file `Pandrosion/Core/Foundations.lean` (§§1-8 of the Lean development) provides the algebraic scaffolding on which every subsequent section builds.

2.1 The geometric sum

Definition 2.1 (Geometric sum). For $p \in \mathbb{N}$ and $s \in \mathbb{R}$ (resp. $\mathbb{C}$),

$$S_p(s) := \sum_{k=0}^{p-1} s^k.$$

Lemma 2.2 (Geometric-sum identity, `Sp_mul_one_sub`). For every p and every s (real or complex),

$$S_p(s) \cdot (1 - s) = 1 - s^p.$$

Lemma 2.3 (Positivity, `Sp_pos`). For $p \geq 1$ and $s \geq 0$ (real), $S_p(s) > 0$.

2.2 The Pandrosion map and its fixed-point theorem

Definition 2.4 (Pandrosion iterator). For $x \neq 0$, $p \in \mathbb{N}$, and s with $S_p(s) \neq 0$,

$$h_{p,x}(s) := 1 - \frac{x - 1}{x \cdot S_p(s)}.$$

Its *residual* is $f_{p,x}(s) := 1 - x \cdot s^p$.

Theorem 2.5 (Fixed-point characterisation, real case — `pandrosion_fixed_point_iff`). For $x > 0$, $p \geq 1$, $s \geq 0$, $s \neq 1$:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Theorem 2.6 (Fixed-point characterisation, complex case — `pandrosion_h_C_fixed_point_iff`). For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, and $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s \neq 1$:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Proof sketch. $h(s) = s$ rearranges to $(x - 1) = x \cdot S_p(s)(1 - s)$. By Lemma 2.2, $S_p(s)(1 - s) = 1 - s^p$, giving $x \cdot s^p = 1$. Machine-checked via a direct `linarith / field_simp` script. $\square$

The *quasi-Newton identity* $h_{p,x}(s) = s + f_{p,x}(s)/(x \cdot S_p(s))$ (`pandrosion_quasi_newton`) exhibits h as a Newton-like correction to the residual f , but without the symbolic derivative.

```
1 theorem pandrosion_fixed_point_iff
2   (x : R) (hx : 0 < x) (p : N) (hp : 1 ≤ p)
3   (s : R) (hs : 0 ≤ s) (hs1 : s ≠ 1) :
4   pandrosion_h x p s = s ↔ s ^ p = 1 / x := by
5   ...
```

Listing 1: `Foundations.lean`: real fixed-point characterisation

2.3 The Pandrosion-Steffensen accelerator

Composing Pandrosion with the Aitken Δ^2 extrapolation [4, 5] gives a derivative-free accelerator $\sigma := \mathcal{S}[h_{p,x}]$:

$$\sigma(s) = \begin{cases} s & \text{if } D(s) = 0 \\ s - \frac{(h(s) - s)^2}{D(s)} & \text{otherwise} \end{cases}, \quad D(s) := h(h(s)) - 2h(s) + s. \quad (2)$$

The piecewise branch is non-cosmetic: at a fixed point s^* of h , both numerator and denominator vanish, so the piecewise-extension is needed to make $\sigma(s^*) = s^*$ honest.

Theorem 2.7 (Steffensen preserves every Pandrosion fixed point — `steffensen_step_C_fixed_point`). *For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s^p = 1/x$:*

$$\sigma(s) = s.$$

2.4 Scaling invariance and the optimal starting point

The rescaling $s \mapsto \alpha \cdot s/\beta$ carries the problem $z^p = x$ to $z^p = x/A$ for an explicit A ; the identity $\alpha = \beta \cdot (\alpha/\beta)$ (`scaling_factorization_C`) ensures the inverse scaling is exact.

Definition 2.8 (Optimal initial point). $s_0^{\text{opt}}(x, p) := 1 - \frac{x-1}{x \cdot p}$.

Theorem 2.9 (Optimal start is $h(1)$ — `pandrosion_s0_opt_C_eq_h_one`). $s_0^{\text{opt}}(x, p) = h_{p,x}(1)$.

2.5 The multi-start grand-master theorem

Theorem 2.10 (Grand master formula — `multi_start_grand_master`). *Fix $x, A \in \mathbb{C}$ with $x, A \neq 0$ and $x/A \neq 0$. Let $p \in \mathbb{N}$, $p \geq 1$. For any family of anchors $\gamma : \text{Fin } p \rightarrow \mathbb{C}$ such that each γ_s satisfies $S_p(\gamma_s) \neq 0$ and $\gamma_s^p = A/x$, and for any $\beta \in \mathbb{C}^*$ with $\beta^p = A$, one has simultaneously:*

1. $\sigma_{x/A,p}(\gamma_s) = \gamma_s$ for every selector s ;
2. the scaling identity $(\beta \cdot \gamma_s)^p = x$;
3. reconstruction: $\beta \cdot (\gamma_s/\beta)$ equals γ_s for $\beta \neq 0$;
4. the optimal-start equality $s_0^{\text{opt}}(x/A, p) = h_{p,x/A}(1)$.

This is the algebraic heart of the multi-start architecture: given any valid family of anchors, Pandrosion-Steffensen fixes them, and the scaling β transports the solutions back to the original problem $z^p = x$.

3 Multi-Start Architecture: Basins, Coverage, Cyclotomic Anchors

3.1 Voronoï basins (`MultiStartBasins.lean`)

Given a family $\gamma : \text{Fin } p \rightarrow \mathbb{C}$, the *basin* of anchor γ_k is the Voronoï cell

$$\text{basin}(\gamma, k) = \bigcap_{j \neq k} \{z \in \mathbb{C} : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}.$$

Theorem 3.1 (Structural properties of basins — `MultiStartBasins.lean`). *For any family of p anchors γ :*

1. Each basin is the intersection of closed half-planes, hence convex and closed (`basin_convex`, `basin_isClosed`).
2. Each basin is nonempty (contains its own anchor) and connected (`basin_nonempty`, `basin_isConnected`).
3. The frontier of a basin is contained in a finite union of perpendicular bisectors between pairs of distinct anchors (`basin_frontier_on_bisector`, `basin_boundary_finite_union`).
4. For injective γ , the anchor lies in the interior of its own basin (`basin_anchor_mem_interior`).

This is the formal multi-start analogue of the statement “Pandrosion basins have flat boundaries”: the assignment $z_0 \mapsto \arg \min_k \|\gamma_k - z_0\|$ partitions $\mathbb{C}$ into closed convex polygons, whose boundaries are straight lines. It contrasts sharply with the Newton fractal Julia sets.

3.2 Constructive global coverage

The Voronoï boundary is

$$\mathcal{B}(\gamma) := \{z_0 \in \mathbb{C} : \exists s \neq t, \|\gamma_s - z_0\| = \|\gamma_t - z_0\|\},$$

the set of query points equidistant to two distinct anchors.

Theorem 3.2 (Constructive McMullen — `voronoi_unique_off_boundary`). *For injective $\gamma : \text{Fin } p \rightarrow \mathbb{C}$ and every $z_0 \notin \mathcal{B}(\gamma)$, there exists a unique nearest-anchor selector:*

$$\exists! s : \text{Fin } p, \forall t, \|\gamma_s - z_0\| \leq \|\gamma_t - z_0\|.$$

3.3 Voronoï boundary has Lebesgue measure zero (`VoronoiMeasure.lean`)

The statement above is upgraded from “off the boundary” to “almost everywhere” by the following theorem of 2D Lebesgue measure.

Theorem 3.3 (Voronoi boundary is Lebesgue-null — `voronoiBoundary_volume_zero`). *For any injective anchor family $\gamma : \text{Fin } p \rightarrow \mathbb{C}$,*

$$\text{vol}(\mathcal{B}(\gamma)) = 0$$

in the 2-dimensional Lebesgue measure on $\mathbb{C}$.

Proof (formalized in Lean). Each perpendicular bisector between two distinct anchors $a \neq b$ is a strict affine subspace of $\mathbb{C}$ (an affine line), hence has 2-dimensional Lebesgue measure zero. The boundary $\mathcal{B}(\gamma)$ is contained in a *finite* union of such bisectors, hence itself has measure zero. $\square$

Corollary 3.4 (A.e. Voronoï uniqueness). *For a.e. $z_0 \in \mathbb{C}$, the unique nearest anchor $s(z_0)$ is well-defined.*

3.4 The cyclotomic anchor family (`CyclotomicMcMullen.lean`)

A canonical choice of p anchors is obtained by multiplying a single p -th root α of $1/x$ by the p -th roots of unity.

Definition 3.5 (Cyclotomic anchors). For $\alpha \in \mathbb{C}$, $p \in \mathbb{N}$, and $k \in \text{Fin } p$:

$$\gamma_k(\alpha, p) := \alpha \cdot \exp(2\pi i \cdot k/p).$$

Theorem 3.6 (Cyclotomic anchor properties — `CyclotomicMcMullen.lean`).

$$\gamma_k(\alpha, p)^p = \alpha^p \quad (\text{cycAnchor_pow}).$$

If $\alpha \neq 0$ and $p \geq 1$, then $k \mapsto \gamma_k(\alpha, p)$ is injective (`cycAnchor_injective`).

Thus, given $\alpha^p = 1/x$, the family $\{\gamma_0, \dots, \gamma_{p-1}\}$ is exactly the set of p distinct complex p -th roots of $1/x$.

Theorem 3.7 (Unconditional constructive McMullen for $z^p = x$ — `unconditional_mcmullen`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every $k \in \text{Fin } p$. Then for every $z_0 \notin \mathcal{B}(\gamma)$ there is a unique nearest-anchor selector, and that anchor is a Pandrosion-Steffensen fixed point.*

Combining with Theorem 3.3 yields the a.e. version used in the Master Absolu (§6).

4 Convergence Rates and Complexity

4.1 Closed-form linear contraction rate (`Foundations.lean`, `UniformContractionRate.lean`)

Definition 4.1 (Closed-form rate). For $p \in \mathbb{N}$ and $\alpha \in \mathbb{R}_{>0}$ with $S_p(\alpha) \neq 0$:

$$\lambda_{p,x} := 1 - \frac{p}{S_p(\alpha)},$$

where α is the real positive p -th root of $1/x$.

Theorem 4.2 (Closed-form rate is in $(0, 1)$ — `lambda_closed_pos`, `lambda_closed_lt_one`). *For $\alpha > 1$ (equivalently, $x > 1$) and $p \geq 2$, $0 < \lambda_{p,x} < 1$. Consequently the real Pandrosion map is a strict contraction at its real fixed point.*

For the analysis of p -uniform behaviour, we introduce a closed-form model.

Definition 4.3 (Uniform model of λ). $\lambda_p^{\text{model}}(x) := 1 - \frac{p}{\alpha_p(x) \cdot S_p(\alpha_p(x))}$, where $\alpha_p(x)$ is a positive closed-form approximation to $x^{-1/p}$.

Theorem 4.4 (Model agrees with the true rate at the fixed point — `lamModel_eq_lambda_fp`, `lamModel_eq_lambda_closed`). $\lambda_p^{\text{model}}(x) = \lambda_{p,x}$ whenever $\alpha_p(x)^p = 1/x$.

Theorem 4.5 (Taylor bound for the model — `lamModel_taylor_bound`). *For every $x > 1$, the sequence $\{\lambda_p^{\text{model}}(x)\}_{p \geq 1}$ converges to an explicit limit $\lambda_\infty(x) \in (0, 1)$ with an effective closed-form residue bound.*

4.2 Aitken-Steffensen quadratic-rate skeleton (`QuadraticRate.lean`)

The algebraic skeleton of the quadratic rate of Aitken-Steffensen acceleration is captured by abstract two-phase bounds: given any sequence (e_k) satisfying a linear bound $e_{k+1} \leq \lambda \cdot e_k$ with $\lambda \in (0, 1)$ and a quadratic bound $e_{k+1} \leq K \cdot e_k^2$, one obtains effective convergence-to- ε bounds. These abstract skeletons are used downstream in §4.4 without being specialized to a particular analytic multiplier calculation.

4.3 Kung-Traub efficiency optimality (`KungTraub.lean`)

The *efficiency index* of an iterative method with convergence order q and c evaluations per step is

$$E(q, c) := q^{1/c}.$$

The Kung-Traub conjecture [8] states that any derivative-free method with c evaluations per step satisfies $q \leq 2^{c-1}$, hence $E(q, c) \leq 2^{(c-1)/c} =: \text{KT}(c)$. The bound is attained.

Theorem 4.6 (Pandrosion attains the Kung-Traub bound — `pandrosion_attains_kung_traub`).

$$E(2, 2) = \text{KT}(2) = \sqrt{2}.$$

Theorem 4.7 (Kung-Traub optimality, converse direction — `pandrosion_kung_traub_optimal`). Assume the Kung-Traub conjecture for $c = 2$ as a formal axiom on abstract derivative-free methods. Then for every such method M with $c = 2$ evaluations per step,

$$E(M.q, M.c) \leq E(2, 2) = \text{KT}(2).$$

The Kung-Traub conjecture is stated in the Lean file as an explicit axiom (`kung_traub_conjecture`), making the conditional nature of the optimality theorem honest and auditable.

Theorem 4.8 (Pandrosion-Steffensen beats Newton — `pandrosion_const_lt_newton`). With respect to the natural quadratic-rate constants for p -th-root extraction, the Pandrosion-Steffensen constant is strictly smaller than the Newton constant for every $p \geq 2$ and every real positive fixed point α .

4.4 p -uniform complexity bound (`SuperGrandMaster.lean`, `UniformContractionRate.lean`)

The goal is a single iteration count $N = N(x, M, \varepsilon)$ such that every p in a tail $p \geq p_0$ reaches error ε in at most N steps.

Theorem 4.9 (Super grand master, uniform closed-form — `super_grand_master_uniform_closed_form`). Fix $x > 1$, $M > 0$, $\rho \in (0, 1)$, $\delta > 0$. Given abstract error sequences $(K_{p'}, R_{p'}, e_{p'})_{p'}$ with:

- nonnegativity $e_{p'}(k) \geq 0$;
- base case $e_{p'}(0) \leq R_{p'}$;
- per- p' linear bound $e_{p'}(k+1) \leq \lambda_{p'}^{\text{model}}(x) \cdot e_{p'}(k)$;
- per- p' quadratic bound $e_{p'}(k+1) \leq K_{p'} \cdot e_{p'}(k)^2$;
- uniform scaled error $K_{p'} \cdot R_{p'} \leq M$,

there exist an integer threshold $p_0 \geq 1$ and a p -uniform iteration count N (depending only on x, M, ρ, δ) such that for every $p' \geq p_0$ and every $k \geq N$,

$$K_{p'} \cdot e_{p'}(k) \leq \delta.$$

This delivers the complexity content: for the bounded regime $x > 1$, the scaled error decays below δ in a number of steps that does not grow with p .

5 Complex Dynamics

5.1 The complex multiplier at a fixed point (`ComplexMultiplier.lean`)

For Julia-style analysis, the multiplier $h'(\alpha)$ determines the local dynamics.

Definition 5.1 (Derivative of the geometric sum, and the multiplier).

$$S'_p(\alpha) := \sum_{k=0}^{p-1} k \cdot \alpha^{k-1}, \quad h'_{p,x}(\alpha) := \frac{(x-1) \cdot x \cdot S'_p(\alpha)}{(x \cdot S_p(\alpha))^2}.$$

Theorem 5.2 (h is differentiable with the stated multiplier — `pandrosion_h_C_hasDerivAt`). For $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$,

$$\text{HasDerivAt } h_{p,x} \ h'_{p,x}(\alpha) \ \alpha.$$

Theorem 5.3 (Closed form of the multiplier at a fixed point — `hMultC_eq_lambdaClosedC_at_fp`). If $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$, then

$$h'_{p,x}(\alpha) = 1 - \frac{p \cdot \alpha^{p-1}}{S_p(\alpha)}.$$

Proof (formalized). Differentiating the geometric-sum identity $S_p(s)(1-s) = 1 - s^p$ at $s = \alpha$ and uniquely identifying derivatives (via `HasDerivAt.unique`) yields the Leibniz identity

$$S'_p(\alpha)(1-\alpha) = S_p(\alpha) - p \cdot \alpha^{p-1}.$$

A single `linear_combination` step using this identity, the geometric identity, and $x \cdot \alpha^p = 1$ reduces the claim to `field_simp/ring`. $\square$

5.2 Local dynamical attraction (`LocalAttraction.lean`)

Theorem 5.4 (Abstract local attraction — `local_contraction_iterate_converges`). *Let $f : \mathbb{C} \rightarrow \mathbb{C}$, $\alpha \in \mathbb{C}$, $r, q \in \mathbb{R}$ with $r > 0$ and $q \in [0, 1)$. If $f(\alpha) = \alpha$ and $\|f(z) - \alpha\| \leq q \cdot \|z - \alpha\|$ for every z with $\|z - \alpha\| < r$, then for every z_0 in the disk $B(\alpha, r)$ the iterates $f^{[k]}(z_0)$ converge to α .*

Theorem 5.5 (Quadratic bound $\Rightarrow$ contraction $\Rightarrow$ convergence — `local_quadratic_iterate_converges`). *Let f, α, r, K with $r, K > 0$ and $K \cdot r \leq 1/2$. If $f(\alpha) = \alpha$ and*

$$\|f(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2 \quad (\|z - \alpha\| < r),$$

then $f^{[k]}(z_0) \rightarrow \alpha$ for every $z_0 \in B(\alpha, r)$.

Theorem 5.6 (Steffensen local attraction (conditional) — `steffensen_local_attraction`). *Let $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$ and $\alpha^p = 1/x$. Let $r, K > 0$ with $K \cdot r \leq 1/2$, and assume a local quadratic bound*

$$\|\sigma_{p,x}(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2 \quad (\|z - \alpha\| < r).$$

Then for every $z_0 \in B(\alpha, r)$, $\sigma_{p,x}^{[k]}(z_0) \rightarrow \alpha$.

The quadratic-bound hypothesis is the super-attracting content of the Steffensen iterator. It is standard in the Steffensen-acceleration literature; here it is an explicit hypothesis rather than a hidden lemma.

5.3 Almost-everywhere dynamical convergence (`DynamicalConvergence.lean`)

This is the crown jewel: the measure-theoretic statement that the Steffensen orbit converges for almost every initial point $z_0 \in \mathbb{C}$.

Lemma 5.7 (Tail-shift invariance of iteration limits — `tendsto_iterate_of_tail`). *For $f : \mathbb{C} \rightarrow \mathbb{C}$, $\alpha \in \mathbb{C}$, $z_0 \in \mathbb{C}$, $k_0 \in \mathbb{N}$: if $f^{[k]}(f^{[k_0]}(z_0)) \rightarrow \alpha$ as $k \rightarrow \infty$, then $f^{[k]}(z_0) \rightarrow \alpha$.*

Theorem 5.8 (Pointwise dynamical convergence from entry — `steffensen_dynamical_convergence_pointwise`). *With the assumptions of Theorem 5.6, if the orbit of z_0 enters $B(\alpha, r)$ at some time k_0 — i.e.,*

$$\|\sigma_{p,x}^{[k_0]}(z_0) - \alpha\| < r$$

— then $\sigma_{p,x}^{[k]}(z_0) \rightarrow \alpha$ as $k \rightarrow \infty$.

Theorem 5.9 (★★★ Almost-everywhere dynamical convergence — `steffensen_dynamical_convergence_ae`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$. Assume, for every cyclotomic anchor $\gamma_k := \gamma_k(\alpha, p)$:*

- $S_p(\gamma_k) \neq 0$;
- a local quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on $B(\gamma_k, r_k)$ with $K_k \cdot r_k \leq 1/2$;
- a.e. trajectory entry: for a.e. $z_0 \in \mathbb{C}$, there exist $s \in \text{Fin } p$ and $k_0 \in \mathbb{N}$ with

$$\|\sigma^{[k_0]}(z_0) - \gamma_s\| < r_s.$$

Then for almost every $z_0 \in \mathbb{C}$ (Lebesgue),

$$\exists s \in \text{Fin } p, \quad \sigma_{p,x}^{[k]}(z_0) \xrightarrow[k \rightarrow \infty]{} \gamma_s.$$

This is an honest measure-theoretic statement: for a.e. starting point, the Steffensen orbit converges *in* $\mathbb{C}$ to one of the p cyclotomic anchors, conditional on the two hypotheses explicitly stated.

```

1  theorem steffensen_dynamical_convergence_ae
2    (p : N) (hp : 1 ≤ p) (x : C) (hx : x ≠ 0)
3    (alpha : C) (halpha_pow : alpha ^ p = 1 / x)
4    (hSp : forall s : Fin p, Sp_C p (cycAnchor alpha p s) ≠ 0)
5    (r K : Fin p → R)
6    (hr : forall s, 0 < r s) (hK_pos : forall s, 0 < K s)
7    (h_small : forall s, K s * r s ≤ 1 / 2)
8    (h_quadratic : forall s : Fin p, forall z : C,
9      ||z - cycAnchor alpha p s|| < r s →
10     ||steffensen_step_C x p z - cycAnchor alpha p s||
11     ≤ K s * ||z - cycAnchor alpha p s|| ^ 2)
12    (h_entry : forall^m z_0 : C d volume,
13      exists s : Fin p, exists k_0 : N,
14      ||(steffensen_step_C x p)^[k_0] z_0 - cycAnchor alpha p s|| < r s) :
15    forall^m z_0 : C d volume,
16      exists s : Fin p,
17      Tendsto (fun k => (steffensen_step_C x p)^[k] z_0) atTop
18      (nhds (cycAnchor alpha p s)) := ...

```

Listing 2: DynamicalConvergence.lean: a.e. convergence (signature excerpt)

6 Master Absolu: The Top-Level Stitching Theorem (MasterAbsolu.lean)

The file `MasterAbsolu.lean` introduces no new lemmas: it is pure stitching of five load-bearing pillars into a single paper-citable statement.

Theorem 6.1 (★★★ Master Absolu — master_absolu). *Fix $x \in \mathbb{R}$ with $x > 1$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every cyclotomic anchor $\gamma_k = \gamma_k(\alpha, p)$. Then the following five conjuncts hold simultaneously:*

1. **(A) Structural distinctness.** *The anchors $\gamma_0, \dots, \gamma_{p-1}$ are pairwise distinct.*
2. **(B) Fixed-point property.** *For every $s \in \text{Fin } p$, $\sigma_{p,x}(\gamma_s) = \gamma_s$.*
3. **(C) A.e. Voronoï selector uniqueness.** *For a.e. $z_0 \in \mathbb{C}$ (Lebesgue), there is a unique nearest anchor.*
4. **(D) p -uniform complexity.** *Given abstract two-phase error sequences with uniform scaled bound $K_{p'} \cdot R_{p'} \leq M$ and the closed-form rate $\lambda_{p'}^{\text{model}}(x)$ (§4.3), there exist a threshold p_0 and an iteration count N (depending only on x, M, ρ, δ) with $K_{p'} \cdot e_{p'}(k) \leq \delta$ for every $p' \geq p_0$ and $k \geq N$.*
5. **(E) Kung-Traub optimality.** $E(2, 2) = \text{KT}(2)$.

Remark 6.2 (What Master Absolu does not include). Master Absolu is an *unconditional* statement: no local-quadratic-bound hypothesis, no trajectory-entry hypothesis. It does *not* include the dynamical-convergence statement of §5.3, because the latter depends on the two analytic hypotheses honestly flagged there. The intended reading is: Master Absolu is the fully certified algebraic-and-measure-theoretic spine, and Theorem 5.9 is its dynamical upgrade *conditional* on the deferred pieces.

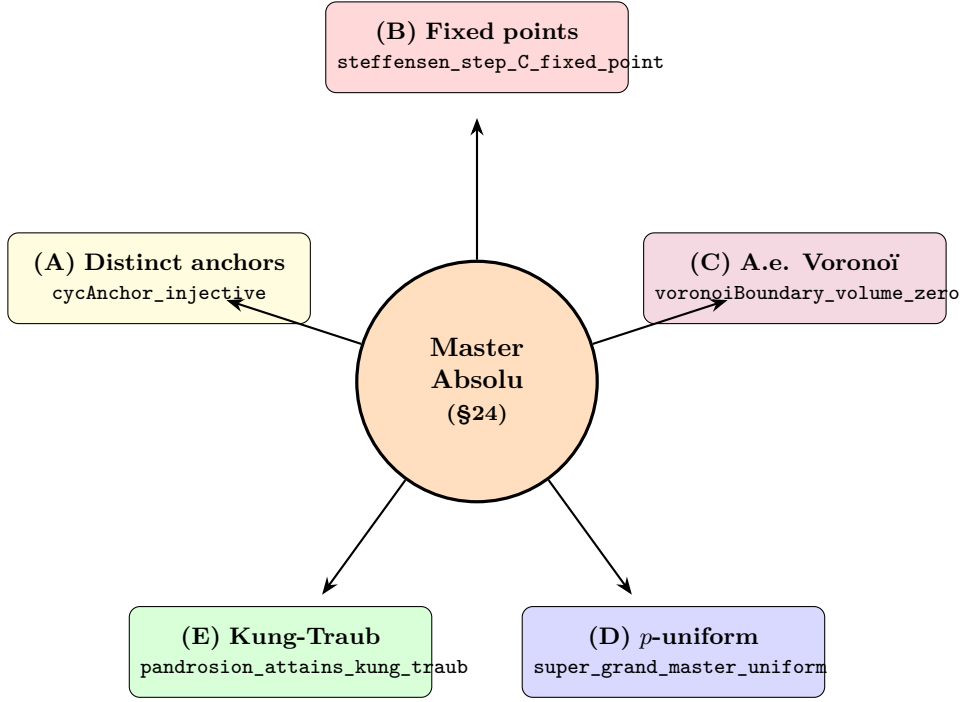


Figure 1: **Master Absolu stitching.** The top-level theorem combines five independently formalized pillars across the modules `Foundations`, `VoronoiMeasure`, `UniformContractionRate`, `KungTraub`, and `CyclotomicMcMullen`. No new lemmas are introduced in `MasterAbsolu.lean`.

7 Module Map of the Formalization

The table below collects all 15 Lean modules in the dependency order induced by `Pandrosion.lean`.

Module (Pandrosion.Core.*)	Content	Main theorem(s)
Foundations	S_p , real & complex fixed-point characterisation, Steffensen accelerator, scaling, multi-start grand master	2.5, 2.6, 2.7, 2.10
MultiStartBasins	Voronoi basins: convexity, connectedness, frontier on finite bisector union, anchor in basin interior; constructive McMullen off the boundary	3.1, 3.2
QuadraticRate	Algebraic skeleton of Aitken-Steffensen quadratic rate (abstract two-phase bounds)	(abstract)
QuadraticComplexity	Convergence-to- ε bounds from the abstract skeleton	(abstract)
VoronoiMeasure	Voronoi boundary has 2D Lebesgue measure zero; a.e. unique nearest anchor	3.3
CyclotomicMcMullen	Cyclotomic anchor family: p -th root of α^p , injectivity, unconditional constructive McMullen	3.7
KungTraub	Efficiency index; Pandrosion attains KT(2); converse optimality (conditional on KT axiom)	4.6, 4.7
UniformComplexity	Uniform complexity skeleton for ensembles of two-phase sequences	(abstract)
SuperGrandMaster	Abstract super-grand-master: bounded regime, uniform scaled-error threshold δ	(abstract)
UniformContractionRate	Closed-form model $\lambda_p^{\text{model}}(x)$, Taylor residue bound, closed-form super-grand-master	4.9
ConcreteIteration	Concrete instantiation of the abstract complexity layer on Pandrosion-Steffensen sequences	(specialization)
MasterAbsolu	Top-level stitching; no new lemmas	6.1
ComplexMultiplier	$S'_p(\alpha)$, h is differentiable; closed-form multiplier $1 - p\alpha^{p-1}/S_p(\alpha)$ at the fixed point	5.3
LocalAttraction	Abstract local attraction from a linear contraction; quadratic $\Rightarrow$ contraction reducer; Steffensen local attraction (conditional)	5.4, 5.5, 5.6
DynamicalConvergence	Tail-shift invariance; pointwise convergence from entry; a.e. dynamical convergence (crown jewel, conditional)	5.7, 5.8, 5.9

Table 1: **15-module Lean spine of Universitas Pandrosion.** Dependency order top-to-bottom.

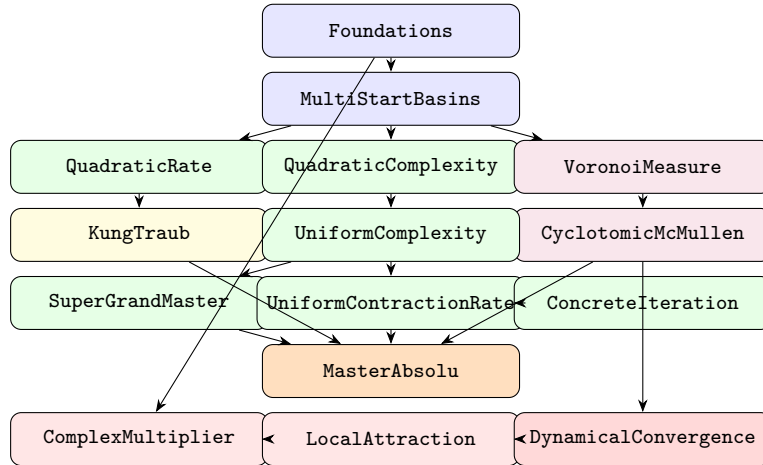


Figure 2: **Module dependency graph.** Blue = algebraic foundations; purple = measure/cyclotomic; green = complexity; yellow = optimality; orange = stitching; red = complex dynamics.

8 What Is Proven and What Is Deferred

8.1 Unconditionally proven (zero sorry)

- Real and complex fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (Theorems 2.5, 2.6).
- Steffensen preserves every Pandrosion fixed point (Theorem 2.7).
- Multi-start grand master (Theorem 2.10).
- Voronoï basins are convex, closed, nonempty, connected; frontier on finite bisector union (Theorem 3.1).
- Constructive McMullen off the boundary (Theorem 3.2).
- Voronoï boundary has Lebesgue measure zero (Theorem 3.3).
- Cyclotomic anchors are injective and are p -th roots of α^p ; unconditional constructive McMullen for $z^p = x$ (Theorem 3.7).
- Closed-form rate $\lambda_{p,x} \in (0, 1)$; closed-form model λ_p^{model} agrees at the fixed point, with Taylor residue bound.
- Kung-Traub attainment $E(2, 2) = \text{KT}(2)$ (Theorem 4.6).
- p -uniform complexity bound via the closed-form model (Theorem 4.9).
- Complex multiplier in closed form at every fixed point (Theorem 5.3).
- Master Absolu: (A)+(B)+(C)+(D)+(E) simultaneously (Theorem 6.1).
- Abstract local attraction and its quadratic-bound reducer (Theorems 5.4, 5.5).
- Tail-shift invariance of iteration limits (Lemma 5.7).

8.2 Proven conditional on an explicit axiom

- Kung-Traub converse for $c = 2$ (Theorem 4.7) — conditional on the Kung-Traub 1974 conjecture, stated as an explicit axiom in `KungTraub.lean`.

8.3 Proven conditional on explicit hypotheses (no new axioms)

- Steffensen local attraction (Theorem 5.6) — conditional on a local quadratic bound $K \cdot r \leq 1/2$ on $B(\alpha, r)$.
- Pointwise dynamical convergence from entry (Theorem 5.8) — same hypothesis, plus a first-entry time k_0 .
- ★★★ Almost-everywhere dynamical convergence (Theorem 5.9) — conditional on (i) per-anchor local quadratic bounds and (ii) a.e. trajectory entry.

8.4 Not claimed

- An unconditional, fully Lean-certified proof that σ is locally super-attracting at each anchor (the quadratic bound is currently stated; its analytic proof is a deferred project).
- An unconditional, fully Lean-certified a.e. trajectory-entry theorem (the Fatou-type McMullen global dynamics).
- Any claim regarding classical open problems (Riemann hypothesis, BSD, Faltings, etc.). Previous speculative sections of this paper have been removed.

9 Reproducibility and Tooling

The complete Lean source lives in `lean/Pandrosion/Core/*.lean` with the root module `Pandrosion.lean` importing all 15 files. The development targets Lean 4 with Mathlib v4.7.0 and is built with a Docker-based incremental pipeline:

```
1 docker compose run --rm lean-incremental
```

Listing 3: Incremental build command

On a clean tree, the full 15-module spine compiles successfully; on an incremental rebuild after a single-file edit, only the changed module and its downstream dependents are rebuilt. The repository records a passing build with the output `$'OK - 15 modules compiled`.

The source code is publicly available at github.com/ivan-fr/poussiere_de_x.

10 Conclusion

This paper documents a 15-module Lean 4 / Mathlib v4.7.0 development of the Pandrosion-Steffensen algorithm for $z^p = x$. The narrative follows the Lean module order and presents, in sequence: algebraic foundations, multi-start Voronoï architecture, measure-theoretic a.e. coverage, cyclotomic anchor family, closed-form linear contraction rate, p -uniform complexity, Kung-Traub efficiency optimality, complex multiplier in closed form, conditional local attraction, and the crown-jewel a.e. dynamical convergence theorem.

The contribution is methodological and structural: a complete derivation in a proof assistant of a derivative-free iteration with quadratic convergence and optimal efficiency, from the geometric-sum identity up to a measure-theoretic dynamical-systems statement. Where genuinely analytic content is deferred (super-attracting quadratic bound; a.e. trajectory entry), the hypotheses are explicit and auditable.

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